

Section 4.2: Homogeneous Linear ODEs — General Solution

Active Recall Study Guide

Part I: Foundations

The Homogeneous 2nd-Order Linear ODE (Source: textbook p.157)

$$ay'' + by' + cy = 0, \quad a \neq 0$$

A solution y must satisfy: y'' is expressible as a linear combination of y' and y . This is a **homogeneous** equation (right-hand side = 0).

The Ansatz: Why y

Exponentials are **eigenfunctions of the derivative operator** D :

$$D(e^{rt}) = re^{rt} \quad \Rightarrow \quad D^n(e^{rt}) = r^n e^{rt}$$

Substituting $y = e^{rt}$ into $ay'' + by' + cy = 0$:

$$ar^2 e^{rt} + bre^{rt} + ce^{rt} = 0 \quad \Rightarrow \quad e^{rt}(ar^2 + br + c) = 0$$

Since $e^{rt} \neq 0$ for all t , we divide out:

$$\boxed{ar^2 + br + c = 0}$$

This is the **auxiliary (characteristic) equation**.

Three Cases by Discriminant Δ

Let r_1, r_2 be roots of $ar^2 + br + c = 0$.

Case 1 ($\Delta > 0$): Distinct real roots $r_1 \neq r_2$

$$y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$

Case 2 ($\Delta = 0$): Repeated root $r_1 = r_2 = r$

$$y(t) = c_1 e^{rt} + c_2 t e^{rt}$$

Why te^{rt} ? Substituting $y_2 = te^{rt}$ into $ay'' + by' + cy$:

$$ay_2'' + by_2' + cy_2 = [2ar + b]e^{rt} + [ar^2 + br + c]te^{rt}$$

When r is a *double* root: $ar^2 + br + c = 0$ (second bracket) AND $2ar + b = 0$ (first bracket, since double root $\Rightarrow \sqrt{\Delta} = 0 \Rightarrow r = -b/2a$). Both terms vanish. (Source: textbook p.163, eq.13)

Case 3 ($\Delta < 0$): Complex roots (deferred to Section 4.3)

Superposition Principle (Source: textbook p.158)

If $y_1(t)$ and $y_2(t)$ are solutions to the homogeneous ODE, then

$$y(t) = c_1y_1(t) + c_2y_2(t)$$

is also a solution for any constants c_1, c_2 .

Proof: Direct substitution and linearity of differentiation:

$$a(c_1y_1 + c_2y_2)'' + b(c_1y_1 + c_2y_2)' + c(c_1y_1 + c_2y_2) = c_1 \underbrace{(ay_1'' + by_1' + cy_1)}_{=0} + c_2 \underbrace{(ay_2'' + by_2' + cy_2)}_{=0} = 0$$

Linear Independence of Two Functions (Source: textbook p.160)

$y_1(t)$ and $y_2(t)$ are **linearly independent** on interval I iff:

$$\neg \exists k \in \mathbb{R} : y_1(t) = k y_2(t) \text{ for all } t \in I \quad (\text{neither is a constant multiple of the other})$$

Equivalently (Problem 33): linearly dependent $\iff \exists c_1, c_2$ not both zero such that

$$c_1y_1(t) + c_2y_2(t) = 0 \quad \text{for all } t \in I$$

Wronskian (Source: textbook p.164)

For differentiable y_1, y_2 :

$$W[y_1, y_2](t) = \det \begin{pmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{pmatrix} = y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

By Lemma 1 (textbook p.160): if y_1, y_2 are solutions to the ODE and $W[y_1, y_2](\tau) = 0$ at *any* point τ , then y_1, y_2 are **linearly dependent** on $(-\infty, \infty)$.

Equivalently: y_1, y_2 are linearly independent $\iff W \neq 0$ everywhere on I .

Existence & Uniqueness (Source: textbook p.159)

For any real $a \neq 0, b, c, t_0, Y_0, Y_1$, the IVP

$$ay'' + by' + cy = 0; \quad y(t_0) = Y_0, \quad y'(t_0) = Y_1$$

has a **unique** solution valid on $(-\infty, +\infty)$.

General Solution Representation (Source: textbook p.160)

If $y_1(t)$ and $y_2(t)$ are **linearly independent** solutions to the ODE on $(-\infty, \infty)$, then $y(t) = c_1y_1(t) + c_2y_2(t)$ is a **general solution**: unique c_1, c_2 can always be found to satisfy any IVP.

The constants solve the 2×2 linear system:

$$\begin{pmatrix} y_1(t_0) & y_2(t_0) \\ y_1'(t_0) & y_2'(t_0) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} Y_0 \\ Y_1 \end{pmatrix}$$

The coefficient matrix is invertible iff $W[y_1, y_2](t_0) \neq 0$, which holds by linear independence.

Part II: General Algorithm

Algorithm: Solving $ay'' + by' + cy$

Step 1. Write the auxiliary equation: $ar^2 + br + c = 0$

Step 2. Compute discriminant: $\Delta = b^2 - 4ac$

Step 3. Find roots: $r = \frac{-b \pm \sqrt{\Delta}}{2a}$

Step 4. Write general solution by case:

- $\Delta > 0$: $y = c_1e^{r_1t} + c_2e^{r_2t}$
- $\Delta = 0$: $y = c_1e^{rt} + c_2te^{rt}$
- $\Delta < 0$: use Section 4.3 (complex roots $\Rightarrow$ trig form)

Step 5. (IVP only) Apply initial conditions:

- Compute $y(t_0)$ and $y'(t_0)$ from general solution
- Set equal to Y_0 and Y_1 ; solve 2×2 system for c_1, c_2

Part III: Problem Solutions

Problems 1–12: Find General Solution

Problem 1: $2y'' + 7y' - 4y$

Step 1. Auxiliary equation: $2r^2 + 7r - 4 = 0$

Step 2. Factor: $(2r - 1)(r + 4) = 0$

Step 3. Roots: $r_1 = \frac{1}{2}$, $r_2 = -4$

Step 4. $\Delta > 0$, distinct real roots:

$$y(t) = c_1 e^{t/2} + c_2 e^{-4t}$$

Problem 2: $y'' + 6y' + 9y$

Step 1. Auxiliary equation: $r^2 + 6r + 9 = 0$

Step 2. Factor: $(r + 3)^2 = 0$

Step 3. Repeated root: $r = -3$

Step 4. $\Delta = 0$, repeated root case:

$$y(t) = c_1 e^{-3t} + c_2 t e^{-3t}$$

Problem 4: $y'' - y' - 2y$

Step 1. Auxiliary equation: $r^2 - r - 2 = 0$

Step 2. Factor: $(r - 2)(r + 1) = 0$

Step 3. Roots: $r_1 = 2$, $r_2 = -1$

Step 4. $\Delta > 0$:

$$y(t) = c_1 e^{2t} + c_2 e^{-t}$$

Problem 6: $y'' - 5y' + 6y$

Step 1. Auxiliary equation: $r^2 - 5r + 6 = 0$

Step 2. Factor: $(r - 2)(r - 3) = 0$

Step 3. Roots: $r_1 = 2$, $r_2 = 3$

Step 4. $\Delta > 0$:

$$y(t) = c_1 e^{2t} + c_2 e^{3t}$$

Problem 8: $z'' + z' - z$

Step 1. Auxiliary equation: $r^2 + r - 1 = 0$

Step 2. $\Delta = 1 + 4 = 5 > 0$

Step 3. Quadratic formula:

$$r = \frac{-1 \pm \sqrt{5}}{2}, \quad r_1 = \frac{-1 + \sqrt{5}}{2}, \quad r_2 = \frac{-1 - \sqrt{5}}{2}$$

Step 4.

$$z(t) = c_1 \exp\left(\frac{-1 + \sqrt{5}}{2} t\right) + c_2 \exp\left(\frac{-1 - \sqrt{5}}{2} t\right)$$

Problems 13–20: Initial Value Problems

$y'(0)$

Problem 13: $y'' + 2y' - 8y$

Step 1. Auxiliary: $r^2 + 2r - 8 = 0$

Step 2. Factor: $(r + 4)(r - 2) = 0$

Step 3. Roots: $r_1 = -4, r_2 = 2$

Step 4. General solution: $y = c_1 e^{-4t} + c_2 e^{2t}$

Step 5. Apply ICs:

$$y(0) = c_1 + c_2 = 3$$

$$y'(0) = -4c_1 + 2c_2 = -12$$

From first equation: $c_2 = 3 - c_1$. Substitute:

$$-4c_1 + 2(3 - c_1) = -12 \Rightarrow -6c_1 = -18 \Rightarrow c_1 = 3, c_2 = 0$$

$$y(t) = 3e^{-4t}$$

$z'(0)$

Problem 18: $z'' - 2z' - 2z$

Step 1. Auxiliary: $r^2 - 2r - 2 = 0$

Step 2. $\Delta = 4 + 8 = 12 > 0$

Step 3.

$$r = \frac{2 \pm \sqrt{12}}{2} = 1 \pm \sqrt{3}, \quad r_1 = 1 + \sqrt{3}, \quad r_2 = 1 - \sqrt{3}$$

Step 4. General solution: $z = c_1 e^{(1+\sqrt{3})t} + c_2 e^{(1-\sqrt{3})t}$

Step 5. Apply ICs:

$$z(0) = c_1 + c_2 = 0 \Rightarrow c_2 = -c_1$$

$$z'(0) = (1 + \sqrt{3})c_1 + (1 - \sqrt{3})c_2 = 3$$

Substitute $c_2 = -c_1$:

$$(1 + \sqrt{3})c_1 - (1 - \sqrt{3})c_1 = 3 \Rightarrow 2\sqrt{3}c_1 = 3 \Rightarrow c_1 = \frac{\sqrt{3}}{2}, c_2 = -\frac{\sqrt{3}}{2}$$

$$z(t) = \frac{\sqrt{3}}{2} \left[e^{(1+\sqrt{3})t} - e^{(1-\sqrt{3})t} \right]$$

Problem 21: First-Order Constant-Coefficient (PRACTICE ANALOG)

Problem 21 — Attempt this yourself first (analog problem)

Do not read ahead until you've attempted Problem 21.

Analog Problem: Consider the first-order linear ODE

$$3y' - 7y = 0$$

- (a) Substituting $y = e^{rt}$, derive the auxiliary equation. What are its roots?
- (b) Write the general solution using those roots.

Once you can do this, apply the exact same logic to Problem 21 in the textbook.

Problems 27–32: Linear Dependence on $(0, 1)$

Test to Apply (Definition 1)

Check: is $\frac{y_1(t)}{y_2(t)} = k$ (constant) for all $t \in (0, 1)$?

- If yes: **linearly dependent**
- If no: **linearly independent**

Alternatively, compute the Wronskian $W[y_1, y_2](t)$.
If $W \equiv 0$ on I : dependent. If $W \not\equiv 0$: independent.

$y_2(t)$

$y_2(t)$

Part IV: Active Recall Summary

Propositional Logic Summary of the Method

Let $P(r)$: “ r is a root of $ar^2 + br + c = 0$ ”

Let Q : “ $y = e^{rt}$ solves $ay'' + by' + cy = 0$ ”

$$P(r) \iff Q \quad (\text{the auxiliary equation is exact iff})$$

Solution space dimension argument (CS framing):

A 2nd-order linear ODE has a solution space of dimension 2 (a vector space over $\mathbb{R}$).

Two linearly independent solutions form a *basis* for this space.

The general solution $c_1y_1 + c_2y_2$ is the *span* of that basis.

IVP = finding coordinates (c_1, c_2) in that basis = solving a 2×2 linear system.

Repeated root analogy (CS): When $\Delta = 0$, we lose one basis vector.

te^{rt} is the second basis vector that restores dimension = 2.

This is analogous to Jordan normal form: a defective eigenvalue requires a generalized eigenvector.

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